

# Siddhant Kumar Upmanyu

Senior Systems Architect · Distributed Systems & Platform

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## PLATFORM ARCHITECTURE & TECHNICAL SCOPE

Senior Systems Architect with end-to-end technical ownership over a commercial IoT smart home and automation platform, with architectural responsibility across cloud infrastructure (Google Cloud Organization Administrator), domain/DNS governance, on-prem bare metal, and production compute clusters. Specialist in low-level Linux kernel networking, high-throughput telemetry storage, JVM runtime optimization, and autonomous SRE triage. Published distributed systems author and open-source framework maintainer.

## TECHNICAL COMPETENCIES

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|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LANGUAGES             | Kotlin, Java, Go, Rust, Zig, C, Shell scripting (POSIX/Bash), SQL                                                                                                                                                        |
| KERNEL & NETWORKING   | Linux kernel ( <code>nftables</code> , <code>iptables</code> ), in-kernel NAT64 ( <code>Jool</code> ), TCP/UDP socket programming, mTLS & custom PKI (CA/CRL), packet reflection, conntrack tuning                       |
| DISTRIBUTED & STORAGE | ClickHouse (columnar modeling, Delta+ZSTD codecs), Redis, PostgreSQL, MongoDB, gRPC, Protocol Buffers, REST                                                                                                              |
| CLOUD & SRE           | Google Cloud (Org Admin, Cloud Run, Cloud SQL, VPC, GCS, IAM), AWS, Terraform / OpenTofu IaC, Docker, Proxmox VE, GitHub Actions, Linux administration (Debian, Ubuntu)                                                  |
| OBSERVABILITY         | Grafana, Loki, Mimir, Alloy, VictoriaMetrics, VictoriaLogs, <code>async-profiler</code> , JMC, VisualVM, Linux performance tools ( <code>sar</code> , <code>vmstat</code> , <code>iostat</code> , <code>pidstat</code> ) |
| METHODOLOGY           | Extreme Programming (XP), London-Style TDD, Consumer-Driven Contract Testing (Pact), Testcontainers                                                                                                                      |

## PROFESSIONAL EXPERIENCE

eGlu Smart Homes (WiZNSystems)

Bengaluru, India · 2022 – Present

Senior Systems Architect (Apr 2026 – Present) | Senior Software Engineer (2024 – 2026) | Software Engineer (2022 – 2024)

### Systems Architecture & Kernel Networking

- In-Kernel Dual-Stack NAT64 ( `Jool` + `nftables` ):** Resolved an edge network partition where cellular IoT hubs operated strictly over IPv6 while backend services were IPv4. Bypassed userspace proxies to preserve CPU credits; deployed the `Jool` kernel module with Explicit Address Mapping Tables (EAMT) alongside `nftables` DNAT to cleanly partition source port ranges on a shared public IPv4.
- Prerouting Packet Reflection & Dynamic Abuse Mitigation:** Offloaded UDP NAT traversal echo responses directly into the Linux kernel using `nftables` packet reflection ( `notrack` ) at prerouting priority, avoiding userspace context switches. Implemented kernel-level DDoS mitigation via `nftables` dynamic sets ( `limit rate over 10/second burst 20 packets` at prerouting priority -301) dropping sweeps before socket allocation.
- Zero-Downtime Kernel Switching:** Designed zero-downtime deployment pipelines using `iptables` DNAT cutovers across both `PREROUTING` and `OUTPUT` chains, explicitly flushing the `conntrack` state table to switch backend traffic without dropping active connections. Hardened deployment and recovery tooling with atomic routing mutations and pre-flight health checks following high-pressure live incident triage.
- Academic Preprint Publication (Zenodo):** Authored and published *Connection-Agnostic Presence Tracking for Stateless Distributed Backends*: formulated a Redis sorted-set architecture with expiration deadlines and throttled batch writes to track IoT device state with mathematically bounded detection latency.

### High-Throughput Telemetry & Storage Systems

- ClickHouse Ingestion Engine (95%+ Compression):** Resolved filesystem inode exhaustion caused by legacy directory log dumps. Designed high-throughput columnar ClickHouse schemas ( `FixedString` , `LowCardinality` , Delta + ZSTD codecs, sparse composite primary keys), slashing telemetry footprint by 95%+ (compressing `device_health` telemetry from 17 GB down to 300 MB on disk). Executed zero-downtime partition-level backfilling and restore operations without slow `INSERT INTO` batches, and devised partition-level disaster recovery backup and restore policies.
- Decoupled Ingestion Microservice:** Engineered an auxiliary Go microservice communicating via binary gRPC with strict Protocol Buffers contracts to encapsulate ClickHouse, shielding the primary Kotlin/Spring Boot monolith from database driver coupling.
- Zero-Downtime Storage Architecture:** Architected and orchestrated an end-to-end multi-phase data migration decoupling binary Base64 image blobs from primary MongoDB collections to GCS with zero downtime; designed dual-write API contracts with document guardrails ( `imageMigrated` ), executed an asynchronous backfill pipeline with immutable GCS caching headers, coordinated backward-compatible mobile rollouts, and purged legacy blobs via `$unset` , slashing database storage by 54x (3.6 GB to 67 MB) and runtime memory by 89% (4.6 GB to 500 MB).
- Production MongoDB DBA & Index Engineering:** Acted as hands-on DBA across heavy-load production MongoDB clusters. Audited slow query logs and execution plans ( `explain` ), pruned redundant/overlapping indexes to reclaim WiredTiger cache memory and reduce write amplification, refactored hot document schemas, and engineered selective compound indexes (strict Equality-Sort-Range ordering) to eliminate collection scans and in-memory sorts under sustained IoT write traffic.

## Concurrency, Runtime & JVM Performance

- **Dynamic IoT-Aware ThreadPool:** Overhauled JVM thread management under bursty IoT packet waves by replacing default thread pools with an adaptive dynamic ThreadPool that scales worker threads based on real-time packet velocity and queue depth.
- **Deep JVM Profiling & Zero-Allocation Hot Paths:** Instrumented live production systems with `async-profiler` and JMC flame graphs. Mitigated allocation churn with buffer pooling and `ThreadLocal` storage; refactored critical packet decoding loops from `String.format` to Java 17 `HexFormat` and bitwise arithmetic, eliminating hot-path CPU bottlenecks. Pinned heap boundaries ( `-Xms = -Xmx` at 4GB) and tuned GC for low pause times.
- **Redis Connection Pool Forensics:** Diagnosed and eliminated silent "ghost connections" and thread pool poisoning in Redis Pub/Sub drivers (Lettuce vs. Jedis) that caused cascading application timeouts under transient network partitions.

## Platform Engineering, Governance & Cloud FinOps

- **Organizational Cloud Governance & FinOps:** Google Cloud Organization Administrator managing IAM policies, VPCs, and corporate DNS zone files. Analyzed multi-year resource utilization to execute a 3-year Google Cloud Committed Use Discount (CUD) on compute instances; currently leading workload evaluation for migrating production workloads from legacy N1 to modern N4 machine types.
- **Production Observability Stack:** Replaced legacy ad-hoc disk logging and custom diagnostic tooling with an enterprise observability stack: Grafana, Loki, and Mimir fed by Alloy with blackbox HTTP/TCP probes, managed under git-tracked configuration (pragmatic GitOps).
- **Zero-Trust Diagnostic Tooling ( `hlogger` ):** Built a secure client-server log inspection utility in Go using Mutual TLS (mTLS) with a custom PKI hierarchy (Root CA, Intermediate CA, client certificates, CRL), eliminating unauthenticated engineer SSH access to production instances.
- **On-Prem Staging Parity & Bare-Metal Homelab:** Built a dedicated on-prem staging environment replicating production network topology, establishing a mandatory pre-production validation gate. Configured a Proxmox VE bare-metal cluster to evaluate container orchestration (Kubernetes CNI/CSI, HashiCorp Nomad) for future platform containerization.
- **Declarative Marketing Infrastructure:** Decoupled the public marketing website completely from core backend infrastructure into its own isolated GCP project, codifying 100% of Cloud Run, Cloud SQL (PostgreSQL), networking, and IAM via Terraform / OpenTofu.
- **Platform Automation & Integrations:** Extended scene/rule automation engines for multi-hub environments (fragment handling, cross-hub synchronization); built cloud-to-cloud OAuth integrations for Yale smart locks, Google Home, and Alexa; rewrote hardware device provisioning flows and replacement operations for failed nodes.
- **Technical Hiring:** Formulated backend engineering evaluation rubrics and practical TDD assessments; interviewed and hired software engineering talent for the platform.

## OPEN-SOURCE SYSTEMS & RESEARCH

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**stopgap** Kotlin · Maven Central ( `2.8.0` )

Microservice framework on Helidon SE (Nima) + Project Loom virtual threads. Custom compile-time DI via KSP (zero runtime reflection). Three-tier testing harness: unit → in-process integration server → Docker E2E via Testcontainers.

**assertgo** Go · Generics

Type-safe testing assertion library built with modern Go generics. Provides a fluent assertion API, chainable `Not()`, custom matchers, and zero external dependencies.

**relay** Zig · Systems

Low-level TCP server in pure Zig with a test-driven approach: raw POSIX socket descriptors, manual memory management without libc runtime dependencies, port binding ( `SO_REUSEADDR` ), and `SIGPIPE` suppression.

**hrh** Rust · CLI

Helm Release Helper CLI — automated declarative Helm deployments with diff previews and atomic rollback guarantees via `cargo install`.

**avoid** Shell / Linux · Distro

Minimal, bootable Linux distribution based on Void Linux engineered for server disaster recovery and lightweight headless appliances.

## EDUCATION

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**Bachelor of Computer Applications (BCA)**

2019 – 2022